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**Paul et al.**

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(54) **PROGRAMMABLE CLIMATE  
CONTROLLER FOR A VEHICLE**

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filed on Oct. 25, 2018, now Pat. No. 11,554,631.

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**B60H 1/00** (2006.01)

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CPC ..... **B60H 1/00971** (2013.01); **B60H 1/00735**  
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(58) **Field of Classification Search**

None

See application file for complete search history.

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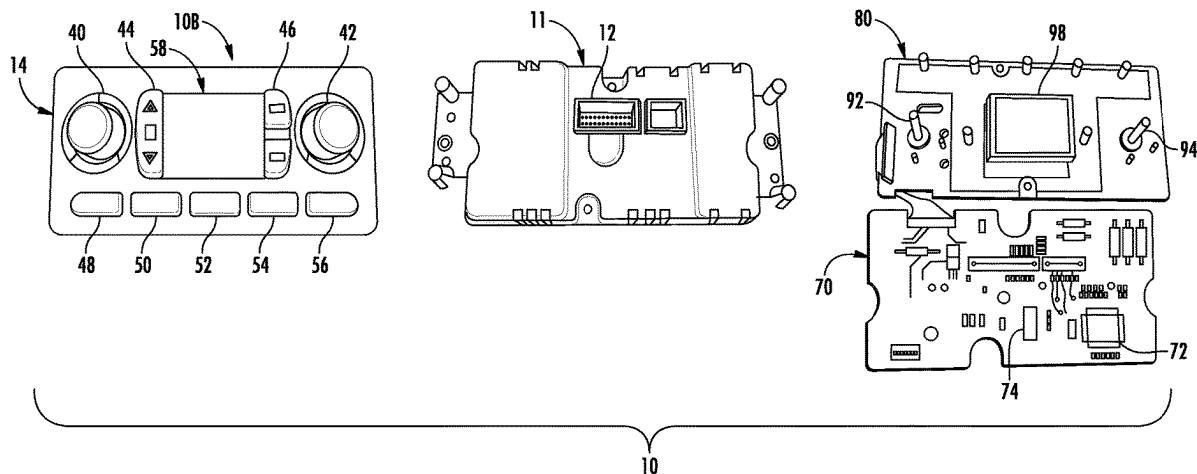
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(57) **ABSTRACT**

A programmable controller stores operational data for a  
number of embedded functions in multiple vehicles from a  
known manufacturer and is programmable to operation in a  
selected vehicle among the multiple vehicles.

**8 Claims, 12 Drawing Sheets**



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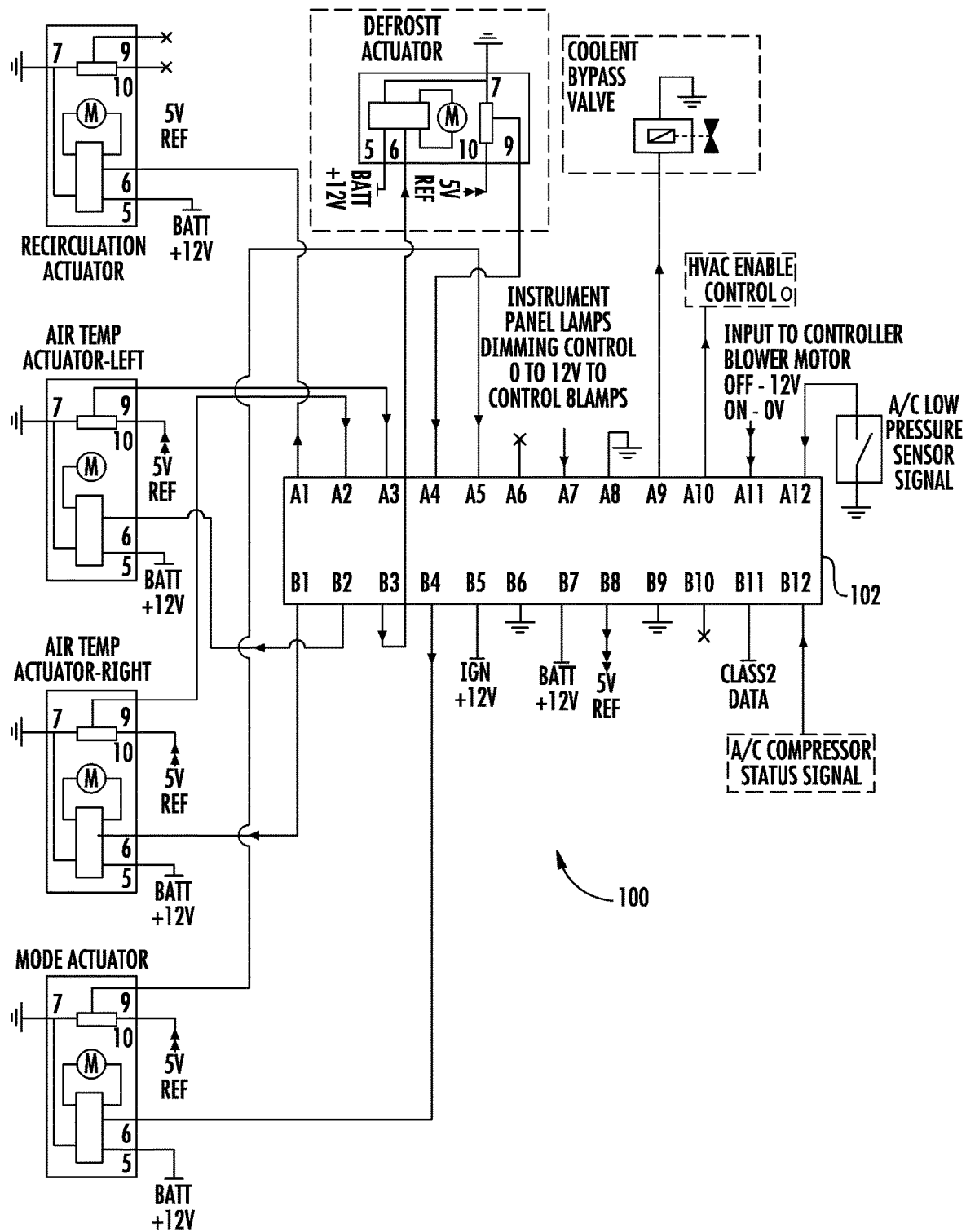
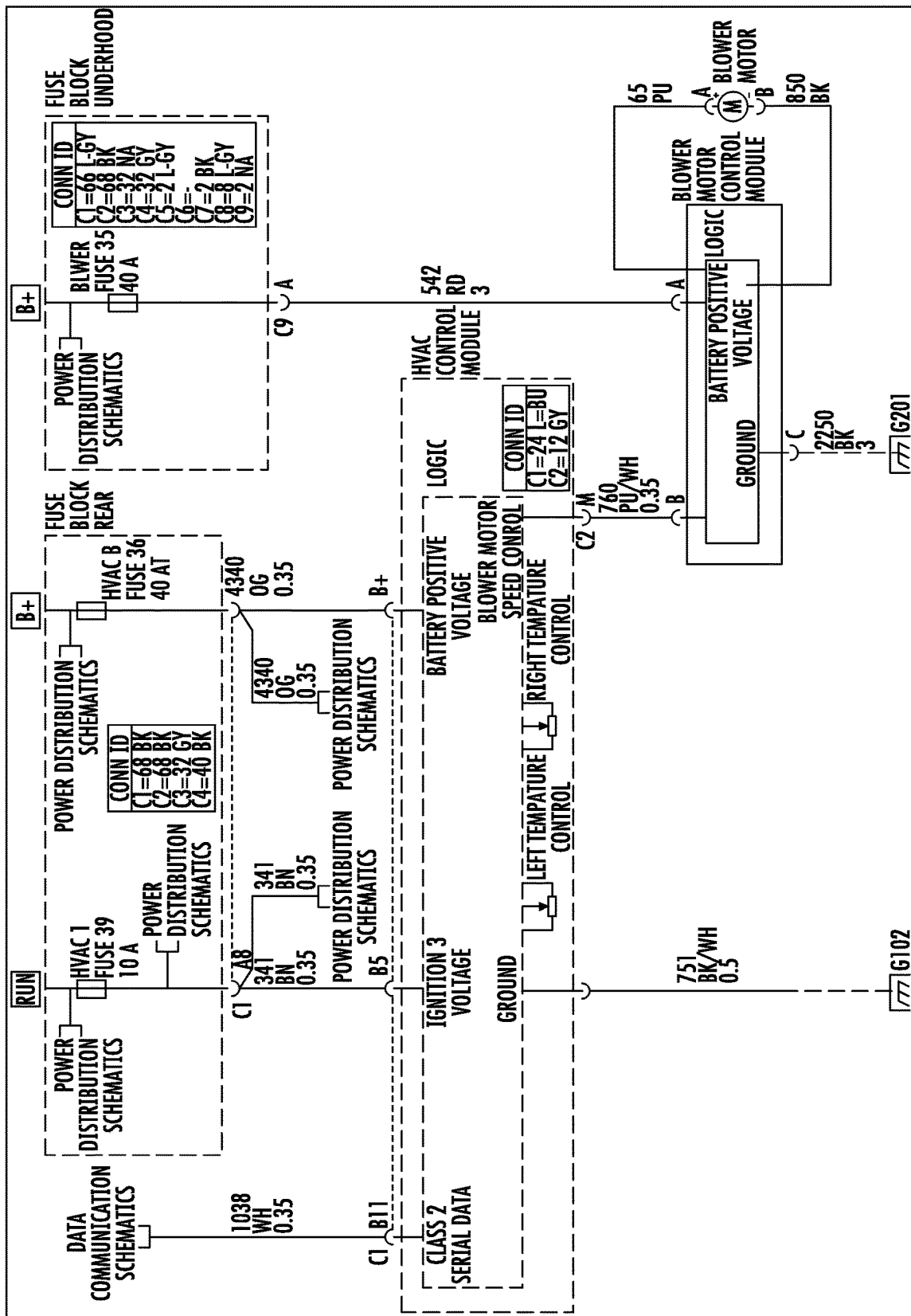
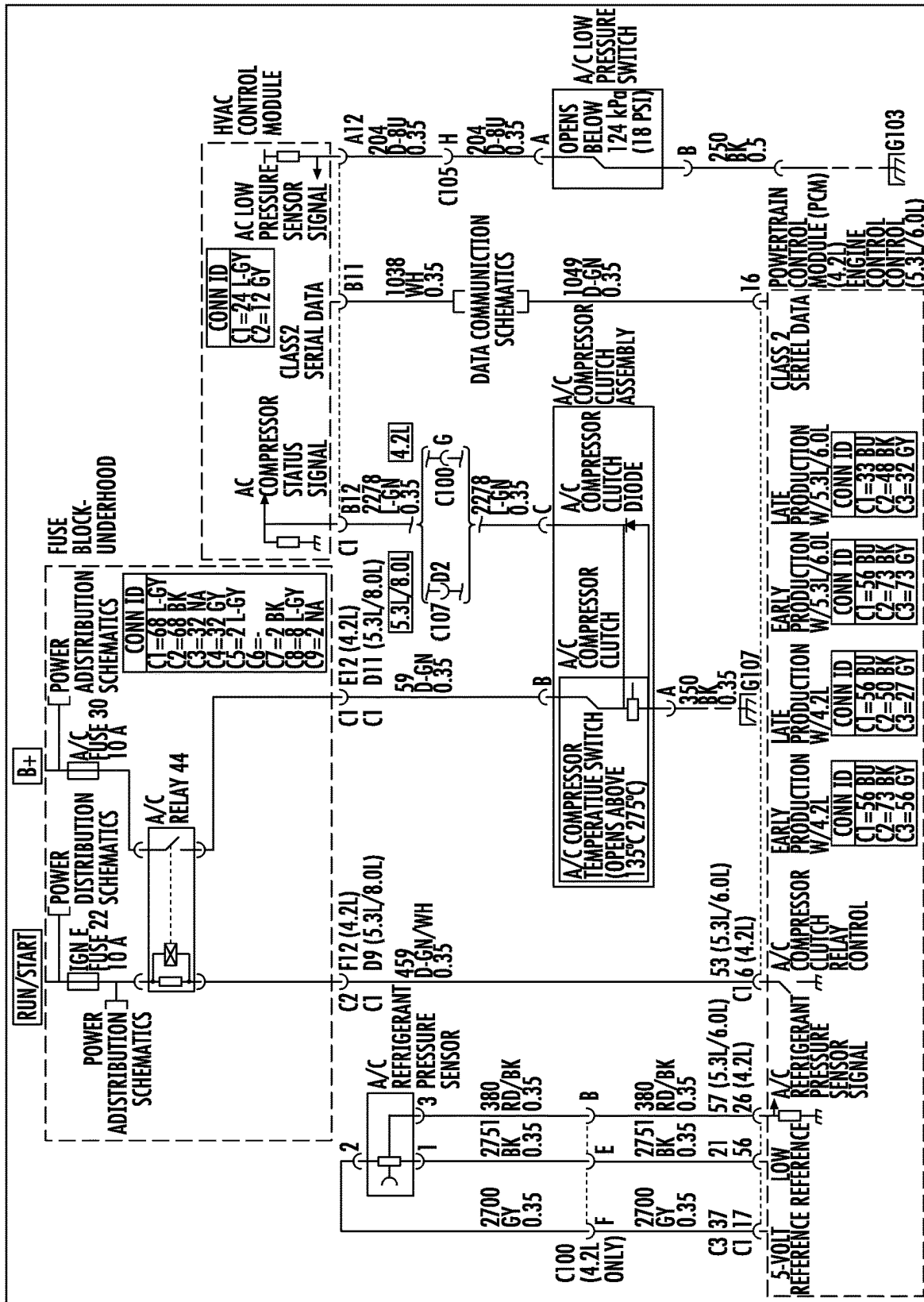


FIG. 1



**FIG. 2**



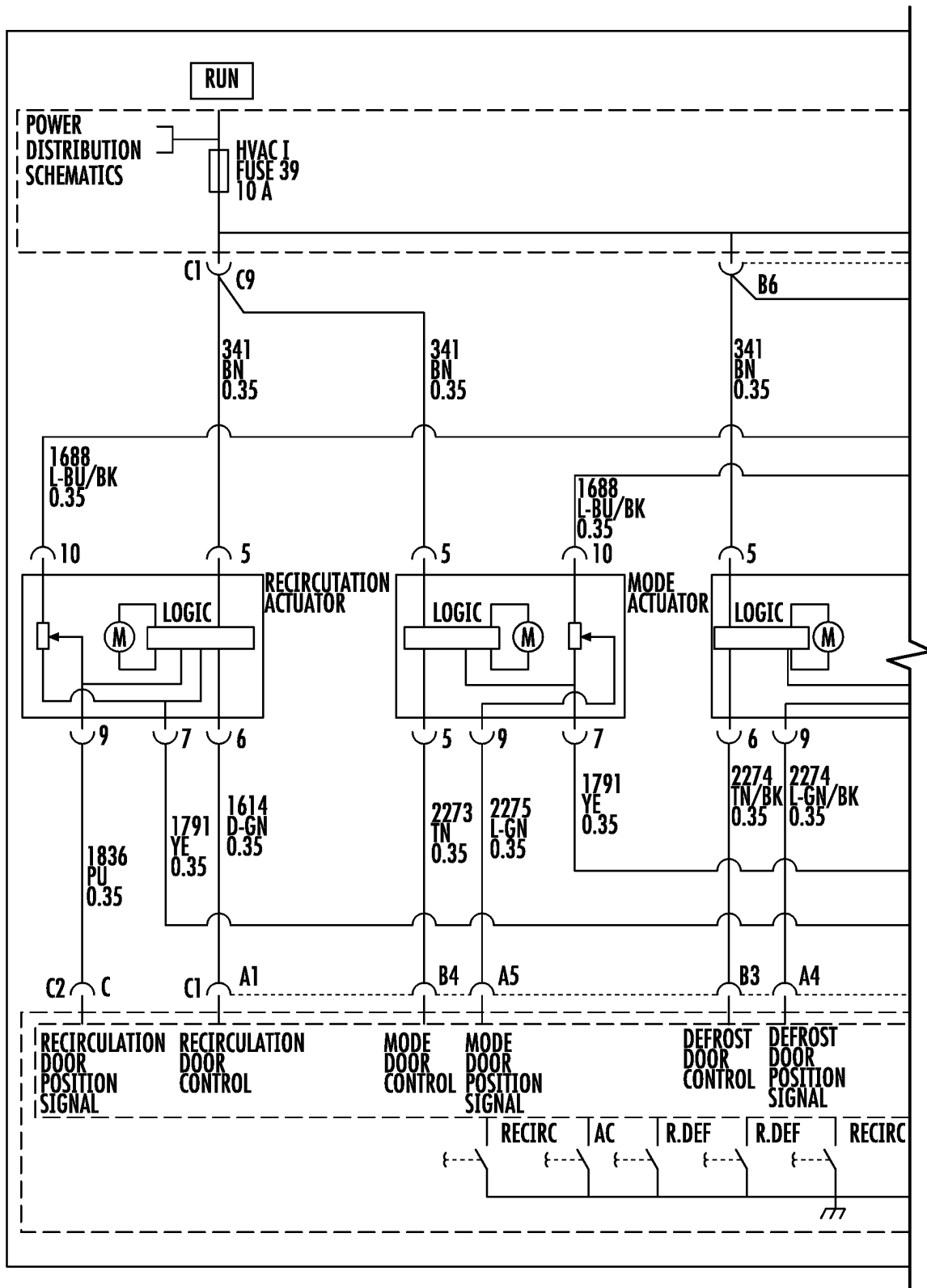


FIG. 4

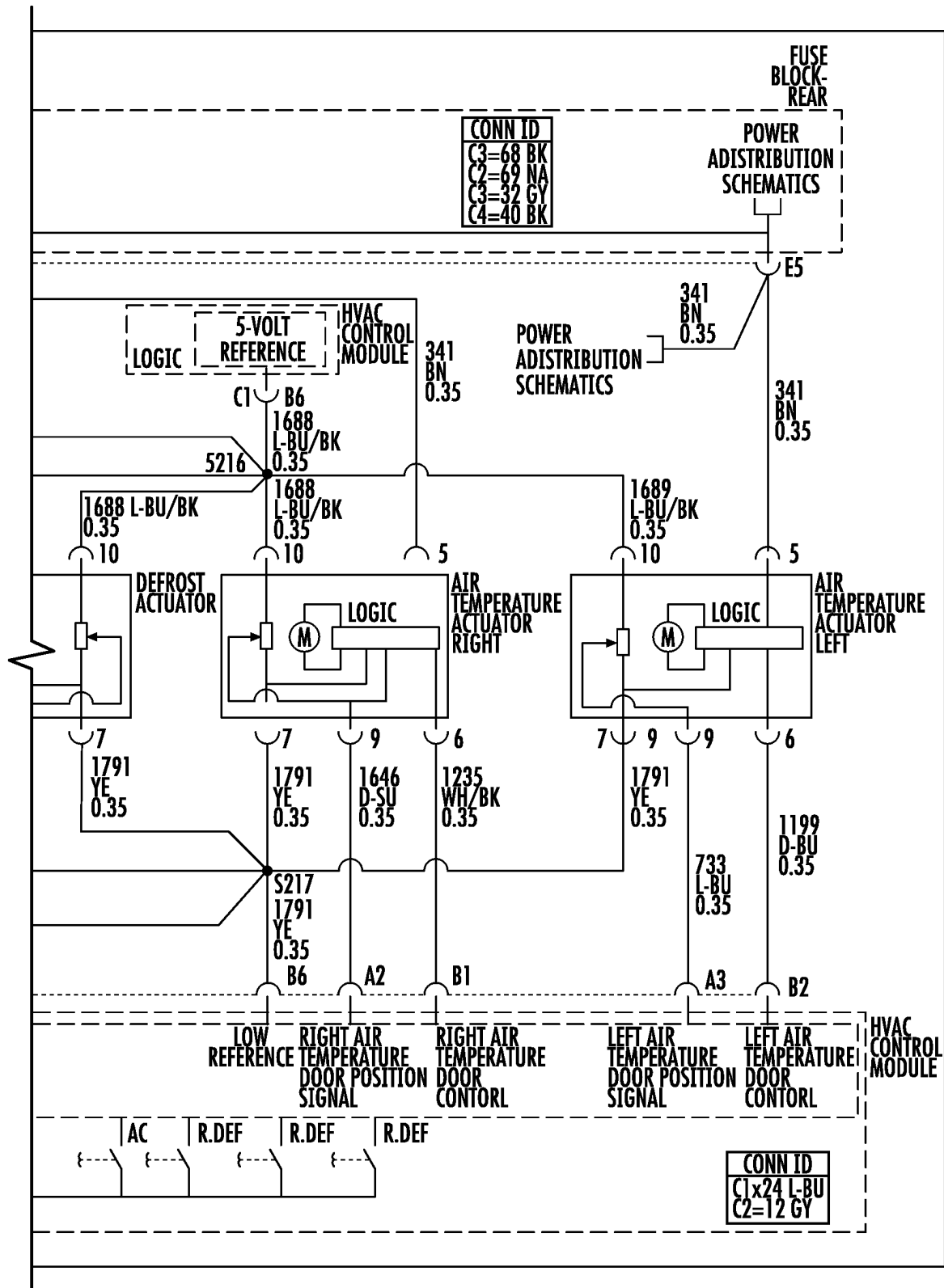
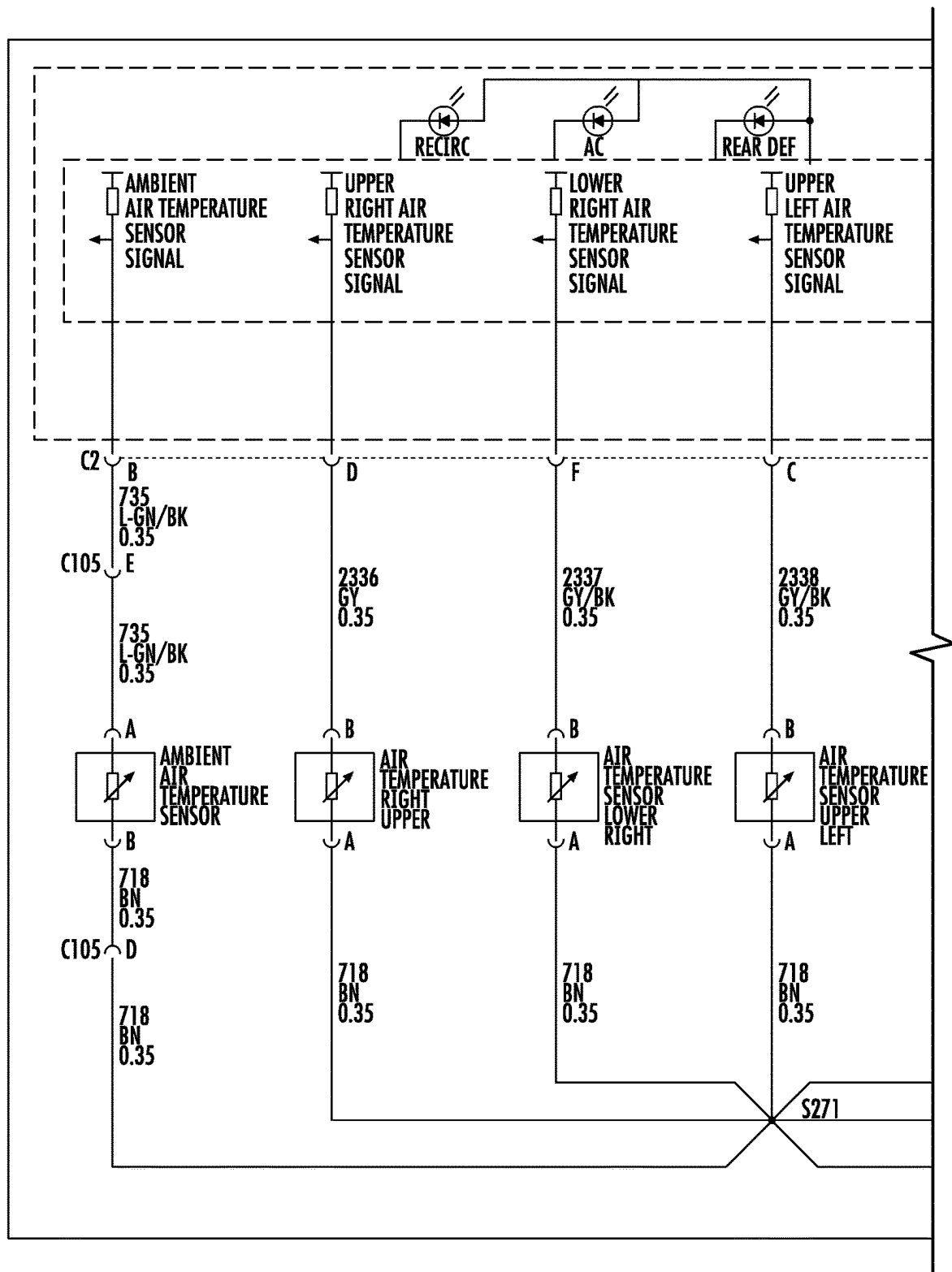


FIG. 4 CONTINUES



**FIG. 5**

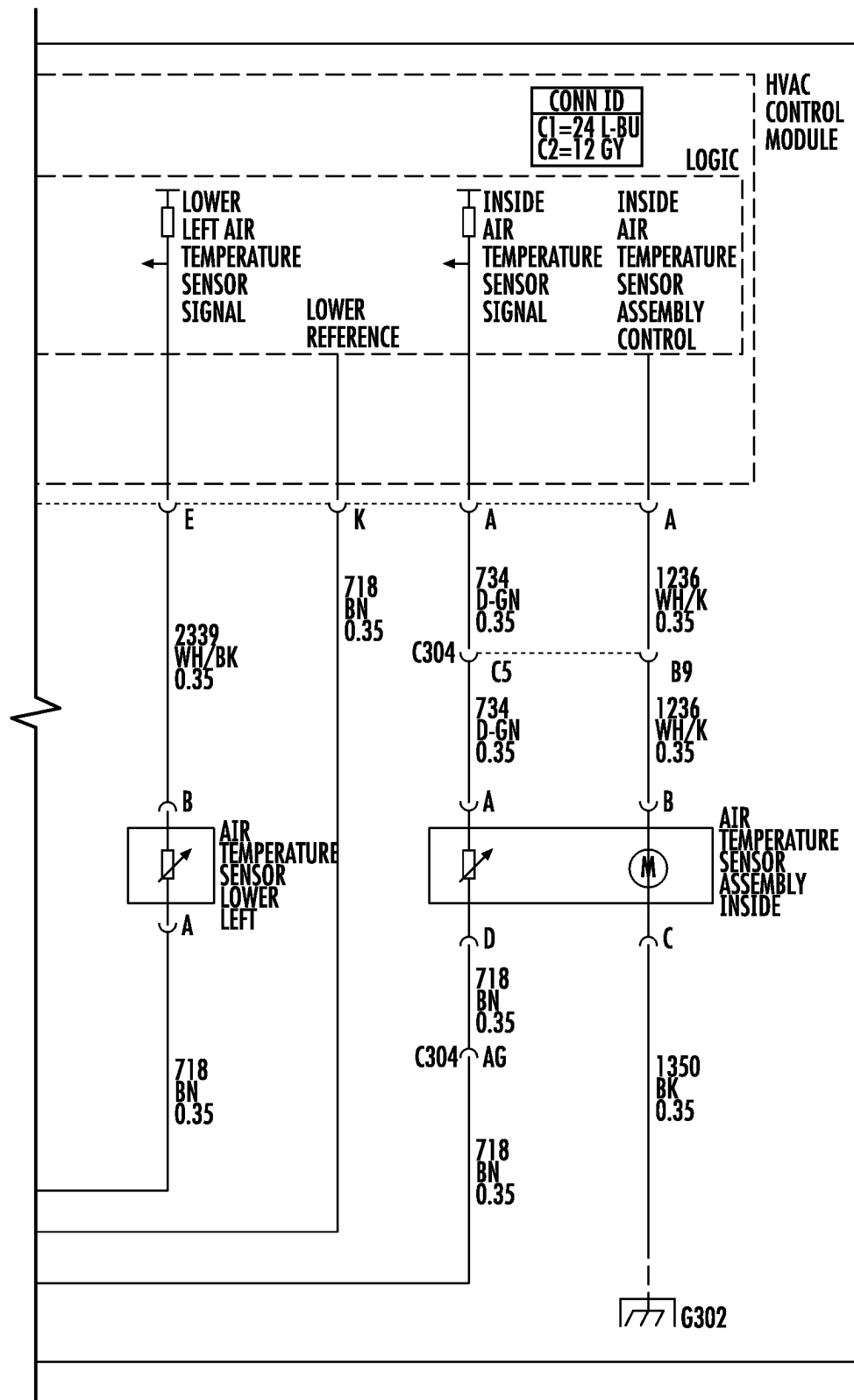


FIG. 5 CONTINUES

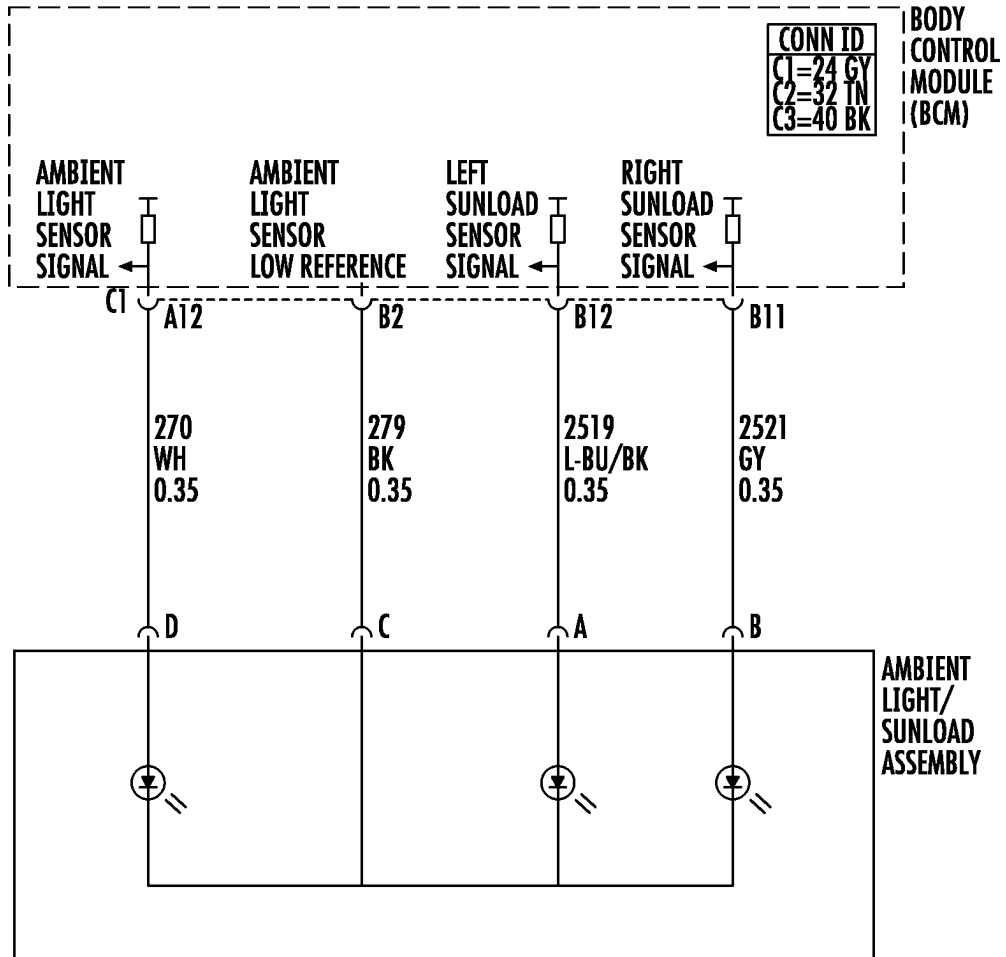


FIG. 6

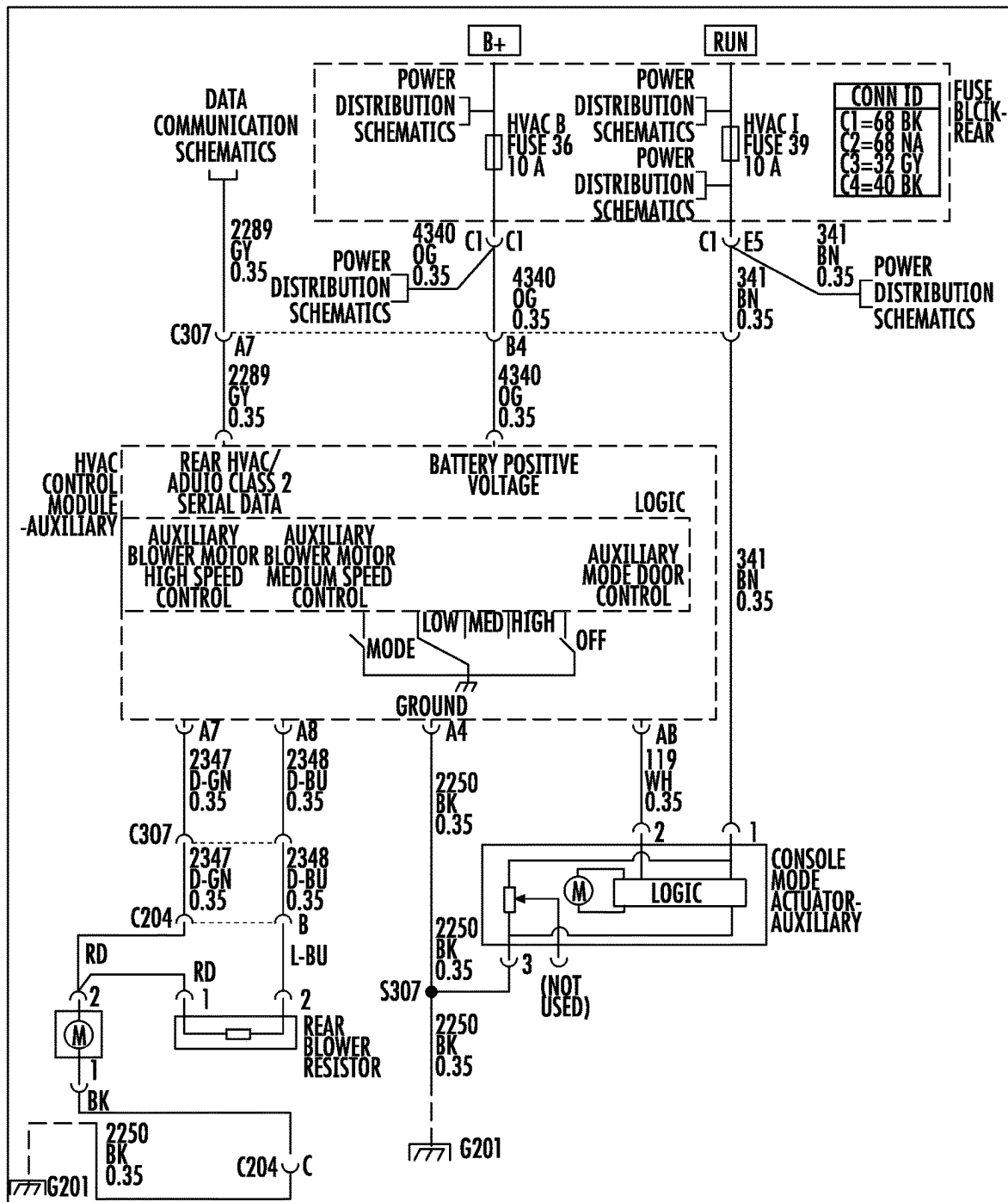
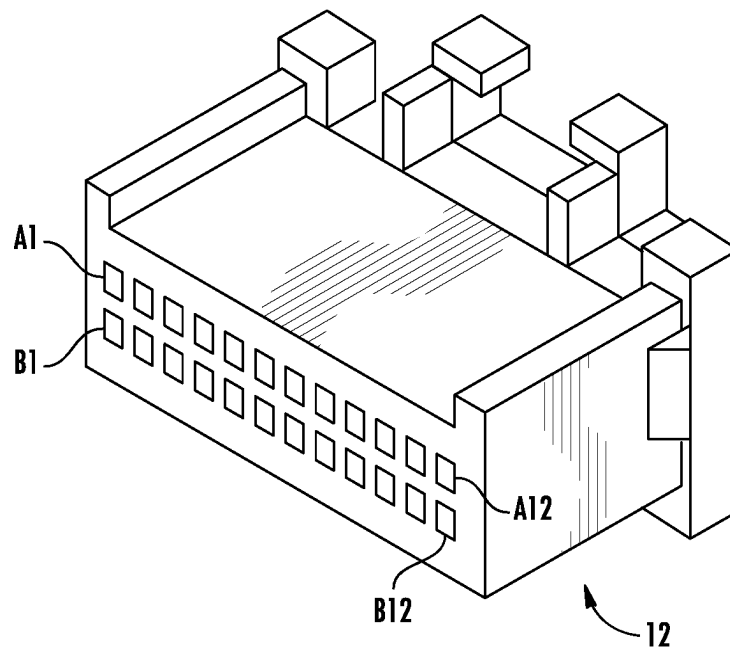
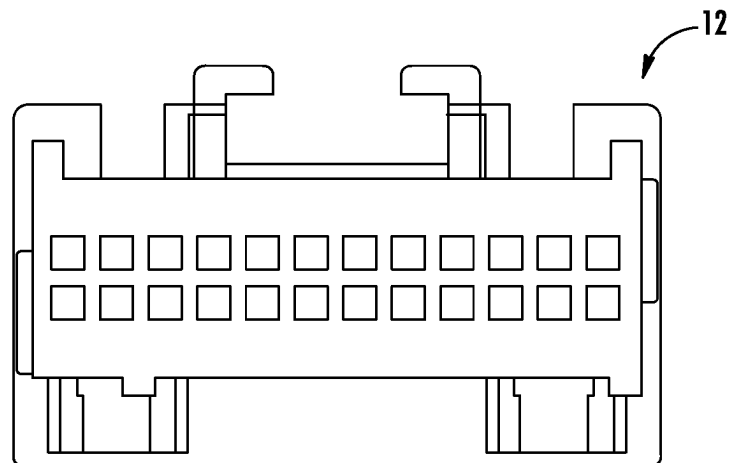


FIG. 7



**FIG. 8A**



**FIG. 8B**

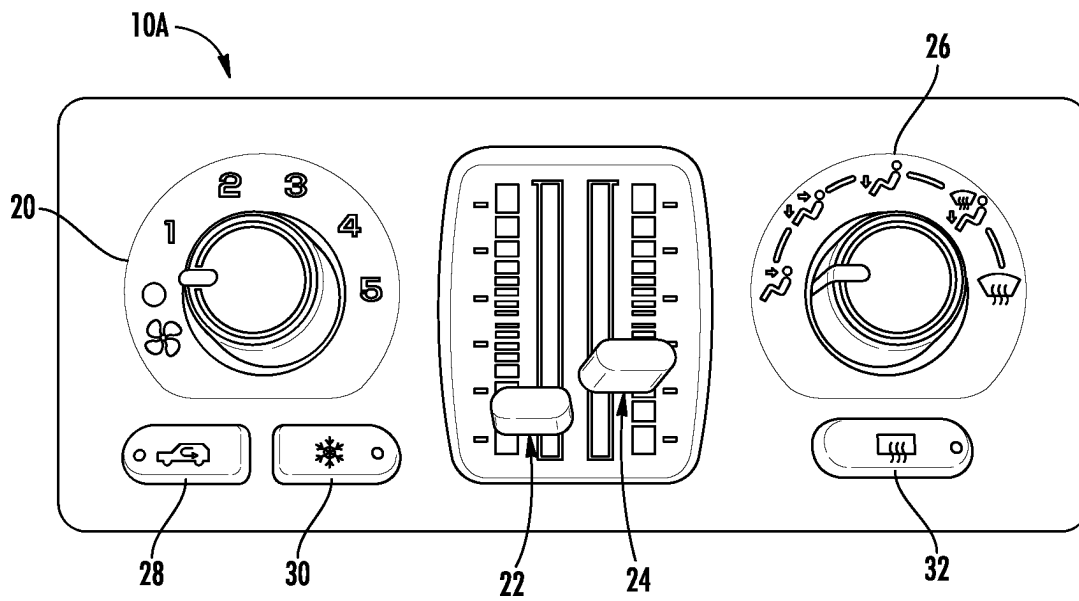
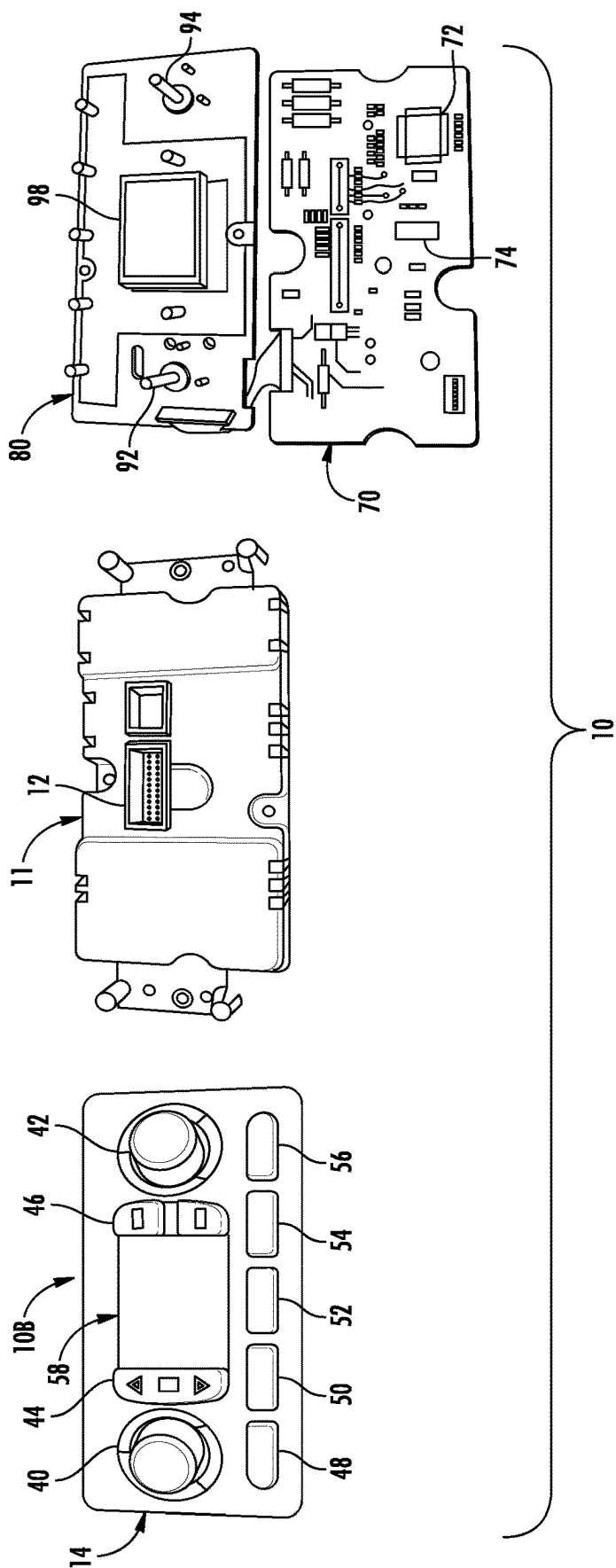


FIG. 9



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## PROGRAMMABLE CLIMATE CONTROLLER FOR A VEHICLE

### CROSS REFERENCES TO RELATED APPLICATIONS

This application claims the benefit of U.S. application Ser. No. 16/866,014, filed on May 4, 2022; U.S. application Ser. No. 16/170,739, filed Oct. 25, 2018; and, U.S. Provisional Application No. 62/577,500, filed Oct. 26, 2017, the entire contents of each is incorporated by reference herein as if fully set forth.

### FIELD OF INVENTION

This invention relates to controls for managing the ambient climate conditions in a vehicle. More particularly, the invention relates to controlling the conditions in the passenger cabin of a vehicle using a heating, ventilation, and air conditioning (HVAC) system. Most particularly, the invention relates to a universal replacement controller that is configured as a replacement for a defective controller in a variety of vehicles.

### BACKGROUND

The ambient conditions in a vehicle are subject to many variables from without and within the vehicle. External conditions, like sun, wind, rain, snow, and frost, influence passage comfort. Internal conditions, like the number and size of passengers, preferences for heating and cooling, compartment size can influence passage comfort. Addressing these ambient conditions is especially difficult when the Original Equipment Manufacturer's (OEM) controller becomes defective or inoperative. While OEM replacement controllers may be available, they tend to be expensive and they can be difficult to find in the marketplace for older vehicles. It is sometimes possible to find used controller on the secondary market, such as at salvage yards; however, the second hand market can be risky and there are seldom guarantees as to their performance.

Moreover, suppliers and retailers have to provide a wide variety of replacement controllers for the OEM variety of different vehicles. Different OEM controllers will typically have variations such as different printed circuit board assembly (PCBA) arrangements and optional components such as rear window defrosters. This variation results in the need for multiple product SKUs with the associated increase in handling, storage, and manufacturing costs.

As a result of these conditions, the marketplace desires a reliable, less expensive option to the OEM replacement controllers. The present invention answers that marketplace need with a replacement controller that is easily programmed according to the specific vehicle application.

### SUMMARY

A method and apparatus for replacing an OEM climate controller is disclosed. A single controller has a number of embedded applications and the selected application within the controller can be called to service through the programming feature of the replacement controller.

In one aspect, the present disclosure is directed to a method for matching a programmable replacement climate controller to a vehicle, including the steps of identifying a vehicle by make, model and year; selecting a programmable climate controller that has embedded climate control data

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compatible with the identified vehicle's existing climate control data and established programming procedures; identifying the location of the embedded data that is compatible with the identified vehicle's existing climate control data; accessing the identified location of the embedded data; and, following the established programming procedures.

In another aspect, the present disclosure is directed to a programmable replacement climate controller comprising embedded climate control data, programming controls for accessing, a connector half, and programming controls that activate. The embedded climate control data is compatible with existing OEM climate control data for a plurality of vehicle groups stored in a memory. The programming controls for accessing are for accessing the embedded climate control data compatible with existing OEM climate control data and extracting embedded climate control data compatible with a selected vehicle group among the plurality of vehicle groups.

### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic illustration of the various features, pin arrangements, parameters associate with a typical OEM controller;

FIGS. 2 through 7 are expanded schematic illustrations of the wiring harnesses that would be associated with the an OEM controller of the type illustrated FIG. 1,

FIG. 8A is a perspective view of a 24-pin connector half for a control module connectable to the half of FIG. 1;

FIG. 8B is a front elevation view of the connector of FIG. 8A;

FIG. 9 illustrates an exemplary dual slide controller face panel; and,

FIG. 10 is an exploded view of an exemplary digital controller face panel, replacement control module, and circuit boards.

### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

With reference to FIG. 1, it can be seen that an exemplary OEM climate control system **100** has a number of designated locations for certain components within the vehicle's OEM wiring harness and those designations terminate on specific pins of the twenty-four pin connector half **102**. In the exemplary climate control system **100** connector half **102** comprises twenty-four pin locations marked A1-A12 and B1-B12 with HVAC components connected to the pin locations. As illustrated in FIG. 1, a Recirculation Actuator has input from A1. A Defrost Actuator has input from B3 and outputs to A4. A Coolant Bypass Valve has input from A9. An HVAC Enable Control has input from A10. Input to Controller Blower Motor outputs to A11. An NC Pressure Sensor Signal outputs to A12. An A/C Compressor Status Signal outputs to B12. A Mode Actuator has input from B4 and outputs to A5. A Right Air Temp Actuator has input from B1 and outputs to A2. A Left Air Temp Actuator has input from B2 and outputs to A3.

The replacement controller **10** has a compatible twenty-four pin connector half **12**, shown in FIGS. 8A, 8B, and **10** that is a direct plug connection to the existing elements of the OEM system **100**. As discussed further below, the replacement controller **10** is configurable by vehicle to have the controller pins functionally compatible with a plurality of OEM climate control systems.

FIGS. 2 through 7 illustrate more detailed representative schematics for the elements of the OEM system **100**. In

particular, FIGS. 2-7 show the electrical schematic diagram for the climate control system of a 2007 Buick Rainier. One of ordinary skill in the art would appreciate that different vehicles have variations in the electrical schematics of their HVAC systems that are nonetheless applicable to a suitable replacement controller 10 as disclosed in the present invention.

FIG. 9 shows the face panel of a replacement controller 10A according to the present invention with a plurality of control elements 20-32 that are used to accomplish configuration and activation in the manner discussed further below and for user operations after the activation programming is completed. Some features, such as rear defrost, are optional on vehicles. In order to provide a single replacement controller 10A applicable to vehicles with or without this feature, the controller can be provided with a variable feature slot that is covered by a removable plate or weakened portion of the cover plate that may be removed without damaging the cover plate. When removed, this variable feature provides a connection point for a functional button 32 that will enable the additional feature.

Because the pin allocations for all target vehicles are not identical and all models may not have all pins operational or active, the replacement unit 10 has a first PCBA 70 (FIG. 10) including a memory module 72 that is preloaded with assignment instructions software for a number of vehicles. Since all models or model years from the same OEM may not have identical features, it is necessary for the software in memory to recognize the target vehicle, the features in that vehicle and the associated pins so that the correct information from the memory software can upload or activate for the target vehicle. It is also important to identify the pin assignments correctly because other features are controlled by the vehicle's on-board computers and the replacement unit 10 must not interfere with the normal operation of any other vehicle features.

The exploded view in FIG. 10 shows a faceplate 14 with digital control elements 40-56 that are used to accomplish configuration and activation as discussed below. The replacement controller 10 also includes a back plate 11 including the connector half 12, a first PCBA or board 70, and a second PCBA or board 90 with control components 92, 94 and a LCD system 98. However, the boards 70, 90 in this exploded view have the same functionality as the earlier faceplate configuration.

The first and second PCBA 70, 90 house the electronic components of the replacement controller 10. The memory 72 stores identifying information for a plurality of vehicles, and may also store configuration and calibration software instructions for a plurality of vehicles. A processor 74 communicates with and controls other components on the first and second PCBA 70, 90, and the processor may contain embedded applications for configuration and calibration of the replacement controller 10. Other electronic components may include network components (including wireless communication components), power components, integrated circuits for implementing any of the above, and the like.

The process of installing the replacement controller 10 begins with identifying the target vehicle in a look-up table and then following the programming steps for the vehicle as shown in a procedure such as the example below. These instructions are applicable to the exemplary digital climate controller 10B shown in FIG. 10 having operative user inputs 40-56 that are typically used for regular operation of the climate controller 10 when fully installed. The particular buttons and displays described below are exemplary, but

other button(s) and displays are contemplated to conform to other OEM components and procedures.

Exemplary Configuration Selection Instructions:

STEP 1: Start the vehicle (the engine must be running).

STEP 2: Enter configuration mode: Press and hold the "Power" 48 and "Front Defrost" 54 buttons until the LCD Screen 58 displays the default configuration number (e.g., "04").

STEP 3: Find the Configuration # for a given vehicle from a reference table such as Table 1.

STEP 4: Select vehicle: Select your vehicle's Configuration Number by pressing the "Fan Up" or "Fan Down" buttons 44. The first digit on the display 58 will change to a "5" (indicating the configuration needs to be set).

STEP 5: Set configuration: Once the desired configuration is displayed, press and release the "recirculation" button 50. The first digit will change back to a "0" (indicates the configuration has been set).

STEP 6: Confirm: Press and release the "auto" button 46 to confirm. The LCD Screen 58 returns to normal operating display information.

STEP 7: Internal calibration: Wait while the system performs a calibration sequence indicated by the recirculation 50 and front defrost 54 button lamps being illuminated. This sequence may take up to, e.g., one minute. Configuration is complete when one or both of these button lamps 50, 54 turn off.

In a representative air delivery system, there are multiple factors with multiple internal variables that must be addressed to provide a replacement unit that will operate in the same manner as the end user has experienced with the OEM unit.

A replacement controller 10 of the present invention will have a plurality of configurations available in memory which are selectable by a user and then executed by a process similar to that above. The replacement controller may have, for example, two to ten such configurations, and each configuration may be applicable to multiple vehicle models and model years.

An exemplary replacement controller 10 with configurations corresponding to different models and years of vehicles is illustrated in Table 1 below.

| Configurations Vehicle Applications |                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01                                  | 2003-2004 Cadillac Escalade; 2003-2004 Chevrolet Avalanche, Silverado, Suburban, and Tahoe; and 2003-2004 GMC Sierra and Yukon.                                        |
| 02                                  | 2007 Chevrolet Silverado; 2007 GMC Sierra; 2005-2006 Cadillac Escalade; 2005-2006 Chevrolet Avalanche, Silverado, Suburban, and Tahoe; 2005-2006 GMC Sierra and Yukon. |
| 03                                  | 2003-2007 Hummer H2.                                                                                                                                                   |
| 04                                  | 2004 Buick Rainier, 2003-2004 Chevrolet Trailblazer, 2003-2004 GMC Envoy, and 2003-2004 Oldsmobile Bravada.                                                            |
| 05                                  | 2005-2006 Buick Rainier, 2005-2006 Chevrolet Trailblazer, and 2005-2006 GMC Envoy.                                                                                     |
| 06                                  | 2002 Chevrolet Trailblazer, 2002 GMC Envoy, and 2002 Oldsmobile Bravada.                                                                                               |
| 07                                  | 2007 Buick Rainier and 2007-2009 Chevrolet Trailblazer.                                                                                                                |

The replacement controller 10 illustrated in Table 1 provides a single replacement device that can replace several OEM controllers. OEM vehicles of a given configuration (e.g., the 2007 Buick Rainier and 2007-2009 Chevrolet Trailblazer) may be considered a "vehicle group" where multiple OEM vehicle types correspond to a single replacement configuration. Thus, a "vehicle group" may be a single

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model in multiple years like Configuration 03 above or multiple models in a single year like Configuration 06 above.

A second exemplary configuration selection procedure is based on holding one or more selected programming elements for a selected time to scroll through the available configurations. A default configuration can be made available based on vehicle popularity and service data and, if the vehicle corresponds to the default configuration, the selection procedure may not be necessary. A second configuration can be selected by: pressing and holding both the Recirculation button **50** and an AC button **52** for five to ten seconds. A corresponding third configuration selection procedure includes: press and hold Recirculation button **50** and the AC button **52** for ten to fifteen seconds. A corresponding procedure for returning to the first (default configuration) includes: press and hold the Recirculation button **50** and the AC button **52** for sixteen to twenty seconds. This configuration selection procedure may have a vehicle state prerequisite such as ignition being off and blower being not in off mode.

FIG. **10** shows multiple views of an exemplary digital programmable replacement climate controller **10B**. The face of the unit **10B**, shown of the left in FIG. **10**, has a number of user inputs **40-56** that enable a user to select a feature and personalize that feature, such as fan speed, temperature, air conditioning, inside/outside air, etc. to the user's preferences. As described above, when the replacement controller **10B** is being installed, the user inputs serve as programming controls. Once the vehicle identification is entered, the inputs **40-56** shift to programming and, after the programming is completed they shift back to user inputs. The connector **12** for the replacement climate controller **10** is shown in the center **11** of FIG. **10**. The circuit boards **70, 90** with the memory **72**, processor **74**, and associated components to achieve compatibility are shown on the right side of FIG. **10**.

The physical characteristics (e.g., size, shape, proportion) of the replacement controller **10** and its constituent parts may be designed to fit within various vehicle dashboards. Although the faceplate **14** is illustrated as substantially planar, it may instead be convexly curved to conform to a particular dashboard shape. Likewise the faceplate **14** may be an oval, trapezoid, or any other shape suitable to replace an OEM controller and fit an OEM dashboard. The length, width, and depth of the faceplate **14**, back plate **11**, and first and second PCBA **70, 90** may likely be modified as dictated by the spatial constraints of the OEM dashboard. In this manner, the replacement unit **10** will provide a suitable fit and finish for the vehicle and will maintain the aesthetic quality of the vehicle interior.

Moreover, one of ordinary skill in the art would recognize that any suitable hardware may be employed for the embodiments described above, particularly the first and second PCBAs **70, 90**. Data, including vehicle identifying information, embedded configurations, and stored instructions, may be stored in the memory **72** of the first PCBA, in a memory on the second PCBA **90**, or in additional storage hardware, permanently or temporarily. Furthermore, the instructions described herein may be implemented in a computer program, software, or firmware incorporated in a computer-readable medium for execution by a computer or processor. Examples of computer-readable media include electronic signals (transmitted over wired or wireless connections) and non-transitory computer-readable storage media. Examples of non-transitory computer-readable storage media include, but are not limited to, a read only memory (ROM), a random

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access memory (RAM), a register, cache memory, semiconductor memory devices, magnetic media, such as internal hard disks and removable disks, magneto-optical media, and optical media such as CD-ROM disks, and digital versatile disks (DVDs). As such, a computer-readable medium, computer, processor, and/or non-transitory computer-readable storage media may be incorporated in any of the components described above, or in additional hardware.

In another embodiment, the invention makes use of the on-board diagnostics connector (OBD II) available in all vehicles after **1995**. The OBD II connector generally has a standard configuration comprised of two parallel rows of pins, 1-8 and 9-16. Vehicle manufacturers are free to select from among various standard communication formats for using the OBD II to communication with their vehicle electronics. As an example, General Motors vehicles typically use SAE J1850 VPW (Variable Pulse Width Modulation) for communications.

In the current example, the J1850 pin is used to communicate between climate controller and the body control module (BCM) and other vehicle electronics. The vehicle identification number (VIN) is retrievable through the J1850 pin and provides useful detailed information about the vehicle's features, such as an air conditioner, a rear defroster, and other features.

Since vehicle specific data is available for virtually all vehicles, the features data for the intended range of vehicles is stored in memory, such as memory **72** of the first PCBA in FIG. **10**. By accessing the VIN through the OBD II connection, it is possible to know the specific list of features associated with a given vehicle. The programmable replacement controller in this embodiment will automatically undergo the configuration procedure after it is connected to the vehicle and secures the VIN. The VIN will then determine the information to be retrieved from memory **72**. The automatic configuration for the target vehicle will begin when the module detects that there is a battery connection and the vehicle ignition is on. The following is an example of the programming procedure.

1. Confirm that the vehicle ignition is in the off position and the vehicle is off.
2. Connect to the OBD II connector half in the vehicle.
3. Start the vehicle and leave it running until the configuration is completed.
4. With the vehicle running, all of the buttons or lights on the replacement module will illuminate to indicate that the vehicle is being identified in the memory.
5. When the vehicle data from memory has been programmed in the module, the illuminated buttons or lights turn off to indicate the programming configuration is completed and the controller is ready for use.

What is claimed is:

1. A replacement controller comprising:

- a memory that stores selectable control data that is compatible with a plurality of vehicles from a known manufacturer;
- a first set of programming controls for selecting data from the memory that is compatible with controller features embedded of an existing on-board control system in a selected vehicle within the plurality of vehicles; and,
- a second set of programming controls for applying the selected data from the memory and rendering the replacement controller operative in the selected vehicle.

2. The replacement controller of claim 1, wherein the selectable control data includes climate control data.

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3. The replacement controller of claim 1, further comprising a faceplate with the first set and the second set of programming controls, and a back plate with a connector half configured to operatively connect with a connector half associated with an existing wiring harness in the selected vehicle. 5

4. The replacement controller of claim 1, wherein the first set and second set of programming controls are user inputs on a faceplate.

5. The replacement controller of claim 1, wherein the replacement controller is contained within a housing and has a connector half with a plurality of pins positioned to mate with selected pins among a plurality of pins that connect with an existing wire harness in the selected vehicle. 10

6. A replacement controller comprising:

15 a memory including control data that is compatible with control systems in a plurality of vehicles from a known manufacturer;

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first programming controls for selecting from the memory control data that is compatible with a control system in at least one vehicle selected from among the plurality of vehicles from the known manufacturer; and, second programming controls for applying the selected controller data compatible with a control system in at least one vehicle and rendering the replacement controller operative in the selected vehicle.

7. The replacement controller of claim 6, wherein the replacement controller has a connector half with a plurality of pins positioned to mate with selected pins in a connector half in an existing wire harness in the selected vehicle.

8. The replacement controller of claim 6, further comprising a face plate with user inputs for operating the first programming controls and the second programming controls.

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